

Ulas Kamaci

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PhD in Electrical and Computer Engineering (UIUC) specializing in machine learning for computational imaging and generative modeling. Thesis on diffusion models and posterior sampling for spectral imaging; separately built the CNN artifact-removal pipeline and data-quality flags adopted into NASA ICON's operational science pipeline.

Skills

Machine Learning: PyTorch, Diffusion Models, Generative Modeling, CNNs, Transformers

Programming: Python, C/C++, Linux, Git

Methods: Optimization (ADMM/ISTA/EM), Unrolled Networks, Bayesian Inference, Uncertainty Quantification

Applications: Computational Imaging, Computer Vision, Signal Processing, Remote Sensing

Education

University of Illinois Urbana-Champaign (UIUC), Urbana, IL
PhD in Electrical and Computer Engineering

Defended July 2026

Middle East Technical University (METU), Ankara, Turkey
BS in Electrical and Electronics Engineering (GPA: 3.76/4.00)

June 2017

Experience

Graduate Researcher, Coordinated Science Laboratory, University of Illinois Urbana-Champaign

Aug 2017 - Jul 2026

PhD Thesis — Learned Reconstruction and Uncertainty Quantification

- Engineered a **deep learning based solver** for a nonlinear imaging inverse problem, trained on **23,000+** patches of Hinode/EIS solar data; **halved** velocity error vs. the classical solver (MART) at **60x** faster inference.
- Derived and implemented **Diffusion Posterior Sampling** with DDPM for the nonlinear inverse problem, reusing a single unsupervised prior across measurement configurations without retraining.
- Built a **deep ensemble** of 8 conditional diffusion models producing per-pixel **90% error bars 2-12x** better calibrated than any single model.

NASA & NSF Mission and Instrument Work

- Built a **CNN-based artifact removal pipeline** for the Far Ultraviolet Imager on the **NASA ICON mission**; adopted by the mission Science Data Center and **deployed in the operational data pipeline**.
- Developed the statistical **uncertainty quantification** method and quality-assessment flags adopted for public release of NASA's **ICON FUV** data to the scientific community.
- Optimized **sensor placement** for a high-resolution **diffractive imaging system** via subset selection, improving reconstruction **SSIM by 32%** over the conventional focal-plane design. (*MAS — NASA Small Spacecraft Technology Program*)
- Integrated a **plug-and-play ADMM** algorithm with the DnCNN denoiser for **multi-frame deconvolution**, improving reconstruction **PSNR by 2.4 dB** (SSIM 0.63 to 0.70) over the STROLLR transform-learning baseline. (*MAS*)
- Designed an efficient **multi-frame image registration** algorithm for highly noisy, motion-blurred satellite measurements. (*VISORS — NSF Space Weather Research Program*)

Teaching Assistant, UIUC

Jan 2021 - Jul 2026

- Delivered graduate-level lectures on deep networks, diffusion models, variational autoencoders, and optimization.

Summer Intern, Aselsan Research Center, Ankara, Turkey

July 2016 - Aug 2016

- Developed a self-supervised sparsifying-transform-learning reconstruction algorithm for **MRI** using the **ADMM** framework.

Selected Publications

- U. Kamaci** and F. Kamalabadi. "Diffusion Posterior Sampling for Slitless Spectral Imaging." *ICASSP 2026*, IEEE.
- U. Kamaci** and F. Kamalabadi. "Solar Slitless Spectroscopy with Learned Reconstruction." *The Astrophysical Journal*, 2026.
- H. U. Frey, S. B. Mende, R. R. Meier, **U. Kamaci**, et al. "In flight performance of the far ultraviolet instrument (FUV) on ICON." *Space Science Reviews* 219, no. 3 (2023): 23.
- E. Widloski, **U. Kamaci**, and F. Kamalabadi. "Optimal measurement configuration in computational diffractive imaging." *ICIP 2020*, IEEE.
- U. Kamaci**, et al. "Efficient sparsity-based inversion for photon-sieve spectral imagers with transform learning." *GlobalSIP 2017*, IEEE.